# ISHWORI KHANAL


Senior Data Engineer

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## Professional Summary


Senior Data Engineer with 5+ years designing cloud-native data platforms across AWS, Azure, and GCP, with deep Snowflake and Databricks expertise. At Visa, automated 120+ enterprise pipelines on AWS (MWAA, EMR, Glue, Redshift, Kinesis), processing multi-million daily financial transactions and driving $180K in annual infrastructure savings. At Kaiser Permanente, architected an enterprise analytics platform on GCP (BigQuery, Dataflow, Cloud Composer) with a Medallion architecture on Databricks and Delta Lake, and built Vertex AI RAG pipelines for natural-language dataset discovery. At Merck, modernized clinical and pharmaceutical workloads onto Azure Databricks, ADLS Gen2, and Delta Lake. Specializes in lakehouse architecture, Spark-native processing, dbt-based transformation, and HIPAA/SOC 2-compliant data governance at scale.


## Technical Skills

- **Cloud Platforms:** Amazon Web Services (AWS): S3, Glue, EMR, Lambda, Redshift, Athena, Kinesis, MWAA, Lake Formation, CloudWatch, IAM, CodePipeline; Microsoft Azure: ADLS Gen2, Azure Data Factory (ADF), Azure Databricks, Azure Event Hubs, Azure Monitor, Azure DevOps, Microsoft Fabric, Azure Cosmos DB, Azure Active Directory (AAD); Google Cloud Platform (GCP): BigQuery, Cloud Storage, Dataflow, Dataproc, Pub/Sub, Cloud Composer, Vertex AI, Cloud Functions, Cloud Run, IAM, Dataplex, Data Catalog, Cloud Build
- **Data Platforms & Warehousing:** Snowflake, Amazon Redshift, Databricks, Delta Lake, Apache Iceberg, PostgreSQL, Microsoft SQL Server, MongoDB, BigQuery, Apache Beam
- **Big Data & Distributed Processing:** Apache Spark (PySpark, Spark SQL, Structured Streaming), Apache Kafka, Hadoop (HDFS, Hive), Apache Iceberg
- **Orchestration & Transformation:** Apache Airflow (MWAA), AWS Glue, dbt, Azure Data Factory, Apache Airflow, Cloud Composer, Dataflow
- **Streaming & Real-Time:** Apache Kafka, AWS Kinesis, Spark Structured Streaming, Azure Event Hubs, Pub/Sub, Dataflow
- **Languages:** Python, SQL, PySpark, Scala, Bash/Shell Scripting
- **DevOps & IaC:** Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, Git, Azure DevOps
- **Analytics & BI:** Power BI, Tableau, Amazon QuickSight, Looker, Looker Studio
- **AI & Machine Learning:** Vertex AI, RAG Pipelines, Vector Search
- **Security & Governance:** RBAC, AWS IAM, Lake Formation, HIPAA, SOC 2, GDPR, Azure Active Directory, IAM, Dataplex

## Professional Experience

**Kaiser Permanente** May 2024 – Present

Senior Data Engineer | Oakland, CA

- Architected scalable healthcare analytics platforms using BigQuery, Databricks, Snowflake, Delta Lake, and Apache Iceberg, enabling governed access to enterprise claims, provider, and member datasets across analytics and reporting teams.
- Led migration of legacy on-prem and SQL Server-based healthcare workloads to GCP, Databricks, and Snowflake, modernizing ingestion and transformation frameworks while improving scalability, resiliency, and operational efficiency.
- Modernized ETL ecosystems by redesigning pipelines with Apache Spark, PySpark, Dataflow, and CDC-based ingestion frameworks, reducing end-to-end processing time for critical healthcare workloads by 43%.
- Identified and resolved performance bottlenecks across distributed data pipelines by optimizing Spark joins, partition strategies, executor configurations, and query execution plans, improving overall pipeline throughput by 38%.
- Reduced data pipeline latency for critical healthcare reporting workflows by redesigning ingestion patterns, implementing Spark caching, optimizing shuffle partitions, tuning Kafka consumer configurations, and enabling parallel processing within Databricks and Dataflow, decreasing downstream data availability delays by 47%.
- Developed distributed batch and streaming solutions using Kafka, Pub/Sub, Spark Structured Streaming, and event-driven processing patterns, enabling near-real-time claims monitoring and operational analytics that improved detection speed and supported timely decision-making.
- Optimized analytical workloads in BigQuery and Snowflake through partitioning, clustering, workload management, and query tuning strategies, improving query responsiveness by 32% for downstream analytics users.
- Built scalable transformation and orchestration workflows using dbt, Apache Airflow, Cloud Composer, and SQL-based transformation frameworks, streamlining dependency management, schema evolution handling, and automated pipeline scheduling.
- Implemented enterprise Medallion Architecture using Databricks, Delta Lake, and reusable transformation frameworks, improving data consistency, lineage visibility, and standardized curation practices across healthcare domains.
- Engineered metadata-driven ingestion, governance, and observability frameworks leveraging Dataplex, Data Catalog, Cloud Monitoring, Spark UI, and Airflow logging, improving operational visibility and reducing production incidents by 27%.
- Designed AI-powered RAG pipelines using Vertex AI, vector search, and enterprise metadata repositories, enabling natural language-based dataset discovery, lineage exploration, and engineering troubleshooting capabilities.

- Automated infrastructure provisioning and platform deployments using Terraform, Docker, Kubernetes, GitHub Actions, and enterprise CI/CD pipelines, reducing environment provisioning efforts from several days to under 4 hours.
- Implemented enterprise-grade security, governance, and compliance controls using RBAC, IAM, encryption policies, HIPAA, GDPR, and secure data sharing frameworks to protect sensitive healthcare information and strengthen audit readiness.
- Collaborated with analytics, compliance, and business stakeholders to deliver trusted healthcare data products while mentoring engineers on Databricks optimization, dbt modeling, Snowflake performance tuning, and modern data engineering best practices.

**Technologies Used:** Google Cloud Platform (BigQuery, Cloud Storage, Dataflow, Dataproc, Pub/Sub, Cloud Functions, IAM, KMS, Dataplex, Cloud Build, Cloud Data Catalog), Apache Spark, Apache Beam, Terraform, SQL, Python, Looker, Looker Studio

**Visa Inc.** Nov 2022 – Apr 2024

Data Engineer | Austin, TX

- Automated orchestration and dependency management for over 120 enterprise pipelines using Apache Airflow (MWAA), improving workflow reliability and operational efficiency.
- Engineered distributed data processing pipelines with Apache Spark, Databricks, and AWS EMR to transform large-scale transactional, customer, and payment datasets across a layered lakehouse architecture, accelerating data availability for downstream analytics.
- Developed scalable ETL/ELT pipelines using PySpark and Snowflake to process multi-million daily transaction and customer datasets through Bronze, Silver, and Gold layer architecture, reducing reporting latency by 33% for fraud analytics and risk monitoring initiatives.
- Optimized storage and compute utilization across enterprise data platforms, contributing to annual infrastructure cost savings of over $180K through workload tuning and archival optimization.
- Designed optimized data models and transformation logic in Snowflake and Amazon Redshift, enabling curated fact and dimension datasets for executive dashboards, customer analytics, and operational reporting.
- Assisted in implementing secure enterprise data access controls using AWS IAM and role-based security models, enabling compliance with governance standards and reducing unauthorized access incidents.
- Monitored distributed pipeline execution, job failures, SLA breaches, and cluster performance using AWS CloudWatch, Airflow Logs, Databricks Jobs UI, Spark UI, and CloudWatch Alarms, improving troubleshooting and operational stability.
- Assisted in building near real-time streaming pipelines using Kafka and Spark Streaming for fraud event ingestion, enrichment, and low-latency analytics processing, reducing event processing delays by 43%.
- Implemented incremental transformation workflows using dbt to standardize raw ingestion data into business-ready curated layers, improving maintainability and enterprise data quality by 31%.
- Developed reusable PySpark, SQL, and dbt transformation frameworks for schema validation, deduplication, enrichment, and standardization of enterprise financial datasets, streamlining data pipelines and improving data quality for downstream reporting.
- Built cloud-based ingestion frameworks to onboard structured, semi-structured, and CDC financial data into centralized data lake and warehouse platforms for downstream analytics consumption, improving ingestion reliability by 37%.
- Implemented automated data quality validations, reconciliation checks, and dbt tests across Bronze to Gold data pipelines to ensure consistency and accuracy of downstream reporting datasets, increasing data accuracy by 28%.
- Participated in CI/CD and deployment activities using Jenkins, Terraform, Git, and Docker, streamlining infrastructure provisioning and release management processes.
- Supported migration of legacy Oracle reporting workflows to modern cloud-based lakehouse and warehouse architectures, improving scalability, governance, and reporting performance.
- Partnered with analysts, compliance teams, fraud teams, and business stakeholders to deliver curated datasets supporting customer behavior analysis, regulatory reporting, and executive-level KPIs.
- Collaborated in Agile/Scrum environments using Jira, Confluence, Git, and pull request reviews, contributing to sprint planning, documentation, code reviews, release tracking, and cross-functional delivery.
- Built scalable lakehouse ingestion patterns using Azure Data Lake and Delta Lake supporting batch and streaming workloads for transaction history, customer activity, compliance, and risk monitoring data domains, enabling faster analytics and reducing data latency.

**Technologies Used:** Amazon Web Services (Amazon S3, AWS Glue, EMR, Redshift, Kinesis Data Streams, Lambda, IAM, KMS, Lake Formation, MWAA), Apache Spark (PySpark), SQL, Python, Oracle, Terraform, Parquet, AWS CloudWatch, Jira, Confluence

**Merck & Co.** Feb 2021 – Oct 2022

Big Data Developer | Rahway, NJ

- Assisted in building batch ETL pipelines using PySpark, Azure Databricks, and SQL to process clinical trial, laboratory, and pharmaceutical sales datasets for enterprise analytics and reporting.
- Collaborated with senior engineers to transform raw healthcare and supply chain datasets into curated Bronze, Silver, and Gold layers using Delta Lake architecture, improving data consistency and reporting reliability.
- Monitored and troubleshot pipeline execution using Azure Monitor, Databricks Logs, Spark UI, and workflow alerts, identifying failures and improving Spark job stability by 23%.

- Assisted in modernizing legacy SQL-based reporting workflows into automated cloud pipelines, reducing manual reporting efforts by 38% and improving operational scalability.
- Supported dimensional data modeling efforts for enterprise reporting dashboards consumed through Power BI, enabling improved visibility into sales, inventory, and operational metrics.
- Supported development of scalable data lake structures using Parquet and partitioning strategies, improving downstream query performance and data retrieval efficiency by 27% across analytical workloads.
- Developed reusable PySpark transformation scripts for data cleansing, standardization, deduplication, and validation of pharmaceutical product and manufacturing datasets, resulting in higher data quality and faster downstream analytics.
- Built parameterized and reusable workflow components to simplify onboarding of new datasets and improve pipeline scalability across business domains.
- Assisted senior engineers with Spark performance tuning, partition optimization, and caching strategies, helping reduce batch processing runtime by 22%.
- Collaborated with QA, compliance, and business teams to enforce healthcare data governance standards, including SOC 2 requirements, ensuring consistent and compliant pharmaceutical datasets across the organization.
- Supported secure enterprise data access implementation using RBAC, role-based permissions, and secure authentication practices to protect sensitive pharmaceutical and clinical data assets.
- Implemented data quality checks, schema validation, and reconciliation logic in Spark pipelines using PySpark on AWS Kinesis streams, identifying and flagging anomalies in production, inventory, and clinical datasets.
- Engineered data ingestion workflows with Apache NiFi and Unix scripts to process structured, semi-structured, and CSV/JSON datasets from enterprise applications and supply chain platforms, enabling timely downstream analytics.
- Worked closely with senior data engineers and analysts to support reporting, supply chain analytics, and operational data integration initiatives across healthcare business units, improving reporting turnaround time by 18%.
- Participated in Agile/Scrum cycles using Jira, Confluence, Git, and Microsoft Teams, leading sprint planning and code reviews that streamlined delivery and enhanced technical documentation quality.
- Documented pipeline architecture, transformation logic, metadata definitions, and data lineage to improve maintainability and onboarding efficiency for new team members.

**Technologies Used:** Microsoft Azure (Data Factory, Azure Databricks, ADLS Gen2, Azure Monitor, AAD, RBAC), Apache Spark (PySpark), Delta Lake, SQL, Python, Power BI, Jira, Confluence, Microsoft Teams, Parquet

**Accenture** Sep 2019 – Jan 2021

Python Developer | Dallas, TX

- Designed and developed scalable backend data processing solutions using Python (2.7/3.x) and advanced SQL, enabling transformation of high-volume operational datasets into analytics-ready relational structures for reporting use cases.
- Engineered complex stored procedures, indexed views, triggers, and performance-tuned queries in Microsoft SQL Server and Oracle 11g, improving query execution efficiency and reducing reporting latency.
- Built reusable batch data transformation utilities using Python (Pandas, NumPy) to cleanse, normalize, and standardize transactional datasets prior to integration into downstream data warehouse systems.
- Implemented structured data ingestion workflows using Apache Sqoop, transferring relational datasets into HDFS to support distributed processing and analytics workloads.
- Developed and optimized large-scale data aggregation pipelines using Apache Hive and MapReduce, enabling efficient processing of multi-million row datasets in distributed environments.
- Created automated file ingestion pipelines using Unix Shell Scripting, streamlining ingestion of CSV and XML vendor feeds into staging and operational databases.
- Designed normalized relational data models following 3NF principles, improving data integrity, consistency, and reducing redundancy across enterprise systems.
- Integrated external vendor systems using Python-based REST APIs, automating secure data extraction and improving operational efficiency.
- Tuned complex SQL views supporting Tableau and SSRS dashboards, optimizing execution plans to improve business reporting performance and responsiveness.
- Supported database migration initiatives consolidating legacy systems into modernized SQL Server environments, improving scalability and maintainability.

**Technologies Used:** Python, SQL, Microsoft SQL Server, Oracle 11g, Apache Hadoop (HDFS, Hive, MapReduce), Apache Sqoop, Unix Shell Scripting, Pandas, NumPy, Tableau, SSRS, Jira, Agile

## Certifications

- Google Cloud Professional Data Engineer
- Snowflake SnowPro Advanced: Architect

## Education

**Atlantis University**

Master's of Science, Information Technology | Miami, FL